TabaPort progress report after IEEE Phase 2

IEEE Response Quest Challenge. Concept 5267.
Flood watch and crew dispatch for Kampala that still works with no internet.

Field	Value
Document	Team research and first build progress report
Date	9 October 2026
Version	3.0
Phase 2	Idea sent 24 May 2026. Moved to Phase 3.
Phase 3	Video of a real person using a working product. Portal closes 9 October 2026.
SRS	First build, October 2026 (docs/SRS.md)

This report puts the team research in one place. Water, drain, map, and software sit with what we then built. **Drawings and site photos come from Eric and the calculation file.** Screen shots come from the first laptop app. News and papers are cited. We do not pretend they are our lab results.

The team

Phase 2 was filed as **one IEEE name** (Shakiran Nannyombi). That PDF did not list a team. After Phase 2, five people did the work. Put these names on the PDF cover, the video end card, and every file you send with the video.

Name	Role	What they did
Shakiran Nannyombi	Lead. IEEE member. Named on Phase 2 and Phase 3.	Idea, firmware, chair of research week, dispatcher on camera, join the parts.
Joel	Water, crew, words on the screen	When a road is unsafe. The 50 percent and 80 percent rule. Job words. Pour and timing on film day.
Eric	Drain, box, sensor place	What drain. How to mount. Trash versus water. Threats. Maths. Hardware. Crew on camera.
Mugisha Trevour Jeanclaude	Maps and land. Makerere Land Surveying. Finish due August 2027.	17 flood places. Earth Engine score. Map files with no internet. Camera and site.
Sharif	Software	Map choice. Four tables in SQLite. USB link. Job flow. SRS. Flask hub.

Working notes also say **Trevor** for Mugisha Trevour Jeanclaude. Add family names for Joel, Eric and Sharif on the printed cover if you have them.

How we split the research (do not do someone else lane):

- Joel: water and the crew. When is a road unsafe? After an alert, who acts?
- Eric: the drain and the box. Where can a sensor sit?
- Trevor: place and the map. Which sites are real? What map is good enough?

- Sharif: software. What can we code in time?
- Shakiran: chair. Write what the four decide. Do not bring a finished design.

1. What this report is for

IEEE Phase 3 needs a **video**. The video is not the research. This file is the research and the build, so a judge or a teacher can see:

1. what Phase 2 claimed
2. what each person found, with sources
3. what we locked as the first build
4. what the product does (with screen shots)
5. what we still do not claim

TabaPort in one sentence: a water sensor on a drain feeds a laptop with no internet. Roads show **green, yellow, red, or unknown**. A job opens so a dispatcher can send a crew **before** the road floods.

It does not stop flooding. KCCA channels move water. TabaPort tells crews **where the water is**.

2. The problem, and the posts we used

When heavy rain hits Kampala, drains at Clock Tower, Bwaise, Kalerwe and other flood spots can close in **minutes**. Plastic and silt block channels. Phone networks jam or drop. Crews, traffic police, boda riders and commuters then cannot see which drain is rising, which road is open, or where to send the next truck. Today that is radio, WhatsApp, and people standing in water.

KCCA is building channels. That does not remove the need for a live water level. Building moves water. TabaPort sits **on top** of that work, also while channels are still being built, and after waste and silt block the new ones.

2.1 Numbers we reuse (as written by others)

Number	What it is	Where we took it
About 362 km of drains	Eight catchments: Nakivubo, Lubigi, Walufumbe, Nalukolongo, Kansanga, Kinawataka, Nakamiro, Mayanja	KCCA briefs. Phase 2. Concept note.
69 flood spots	Sites that still need watch	KCCA. Concept note.
UGX 768 billion	Drain works ask, 2026/27 to 2030/31	KCCA and Parliament briefs
About US$50 million a year	Flood damage (UNDP figure in the inquiry)	Monitor and Parliament story
980 m of Lubigi not finished. Nsoba and Kiwunya just started. Jugula more than 90 percent taken by buildings.	Works are a five year plan, not a finished city	KCCA and UBC updates

KCCA Deputy ED Benon Kigenyi: *"We cannot stop the rain, but we can prepare Kampala to manage it better."* He also said new drains fail if waste and silt keep blocking them (KCCA news 1154. UBC, 12 August 2026).

Parliament asked KCCA for a 7 day report after visits at Nalukolongo Najjanankumbi, Lugogo and Kyambogo.

NTV said KCCA asked for land so channels can grow. Those roads stay small for years. Those are the sites that need a sensor and a detour. Do not say the civil works already solved them.

KCCA is building	TabaPort adds
Bigger channels and culverts	Live water level at those same places
69 flood spots watched by patrol	A ranked job list so the first truck goes to the worst site
Trucks ready when rain starts	A map with no internet: red, yellow, green
Waste and silt as the leftover failure	A median filter for floating plastic, plus a report from a rider who saw the block
A five year master plan	Eyes during the years of building

3. What Phase 2 claimed, and what we cut

Phase 2 file: docs/phase2/QM_Idea_5267_08_23_2026.pdf
Title: Tabaport: On Site Edge Intelligence and Dispatch System for Urban Flash Floods (5267)
Author on the form: Shakiran Nannyombi. Sent 24 May 2026. **Is this a team submission? No.**
Result: moved to Phase 3.

3.1 The idea (seven pages)

1. **Nodes.** ESP32 plus a sealed ultrasonic in culverts. A moving median on the chip so floating plastic is not a flood. Sample every 30 seconds. Aim under US$55 per node.
2. **Hub.** Laptop or Raspberry Pi. SpatiaLite and OSM. No internet.
3. **AHP rank.** Rank sites by how fast water rises, hospitals and flyovers, and how many people live there.
4. **A star routes.** A red road gets a huge cost. Detour on high ground (for example the Northern Bypass).
5. **Human reports.** USSD or SMS from boda riders. Place name match. 100 m circle until a sensor or a crew confirms.
6. **Privacy.** Drop the phone number and IMSI at the door. MIT licence.
7. **Public loop.** Cell broadcast SMS in a 2 km ring and KCCA social posts when a node goes red.

Phase 2 sources named on the form: KCCA Drainage Master Plan. OpenStreetMap Uganda. UNMA. IEEE Systems Council notes.

IEEE review (IT, legal, finance names in the PDF) passed the **idea**. That is not a lab test of the hardware.

3.2 What the first build actually is

Phase 2 line	First build (this report and the video)
Solar, LoRa, BLE, GSM fallback	USB serial, 115200, one cable
SpatiaLite	SQLite and GeoJSON. Map work done **before** the rain
Sample every 30 s	2 Hz (500 ms) so the chart moves on camera. Field sample time is still an open power question
Cell broadcast 2 km	**Not ours.** UCC and the phone companies. A typed report on the hub instead

Phase 2 line	First build (this report and the video)
Raspberry Pi city hub	One laptop. WiFi **off**
US$55 production node	USB lab or campus node (about UGX 138000 to 198000 parts)

Sharif last line, which the team locked: *if we cannot acknowledge, assign and close, we only have a graph, not a dispatch tool.*

4. One system, five owners

The four research files stay one product if they share names. Do not type the same place twice with two different numbers.

ID	Meaning	Owner
place_id TP 01 to TP 17	Flood place list	Maps
segment_id for example clock_tower_01	Coloured road stretch	Sharif and GeoJSON
node_id for example drain_01	Physical sensor	Firmware and hub

Eric owns the culvert and the node hardware. Sharif owns readings, segments and jobs. Joel and Shakiran own crew words and the IEEE cut. Trevor owns the place list and the map files.

5. Joel: water, bands, and what happens after the alert

Source: docs/phase3/thresholds-and-words.md and the research week brief.

Joel was asked not to wait to be told the centimetres. Books do not give one "unsafe for a boda" number for every Kampala drain. Depth is measured **at the site**.

5.1 Two tapes, then the bands

- **H_sensor:** probe face to drain floor (invert), dry.
- **H_road:** road or path edge to invert.

Water depth on the laptop is the same formula Eric drew (Figure E9): H_water = H_sensor minus d_measured.

Band	Default rule	Tank fallback (drain under 40 cm)	Meaning for the road
Green	depth under 50 percent of H_road	under 8 cm	Drain is working. Nothing to do.
Yellow	depth at or above 50 percent of H_road	at or above 8 cm	Filling faster than it empties. Inlet is probably blocked. Send someone **now**.
Red	depth at or above 80 percent of H_road, **or** within 10 cm of the road edge	at or above 15 cm	Water on the road in minutes. Close the lane. Clear the inlet.
Unknown	no reading for 10 s	same	We do not know. **Never green.** Treat as yellow for routes.

Example: 60 cm drain, probe 50 cm above the floor. Yellow at **30 cm**. Red at **48 cm** (also within 10 cm of the 50 cm edge). Settings works this out from H_road.

Why 50 and 80. KCCA drains have freeboard. Half full during rain is fine **only if the water is moving**. Half full and rising means the inlet or outlet is blocked. Eighty percent leaves about the time a city crew needs to reach a flood spot from the Clock Tower depot in traffic (about 10 to 15 minutes) before the path is wet.

Rise rate (cm per minute over the last 60 s) is **shown** in the first build. It does not change the band until we have real storm data. A rise above 5 cm per min in a storm is a later rule that the inlet is blocked, not just a full drain.

5.2 After the alert: who does what

If this flow is missing, Sharif said, we only have a graph.

Trigger	Job words on the laptop
Yellow	**Send drainage crew to check inlet**
Red	**Ask traffic police to close lane. Send drainage crew to clear inlet**
Human report, no sensor	**Send drainage crew to check inlet** (segment yellow. Never red without a sensor or a crew on site)
Opened by hand	dispatcher types the action

Statuses: open, then acked, then en route, then on site, then closed. Side exit at any step: false alarm with a reason.

Rules Joel wrote (and Shakiran says on camera):

1. Acknowledge within **3 minutes**. After that the row flashes.
2. Assign a crew before or at acknowledge.
3. **Closing a job does not paint the road green.** Only the sensor does. If the crew clears the inlet, the water drops and the sensor turns it green itself.
4. False alarm reasons we expect: debris under the probe, crew found the drain clear, test pour.

Demo crews: Team A Central. Team B Kawempe. Team C Rubaga. Traffic Police.

Spoken lines (video):

1. Yellow. *"Campus drain is at thirty centimetres and rising. That is yellow. I am sending Team A to check the inlet."*
2. Red. *"Forty eight. Red. Water is ten centimetres from the path. I am asking traffic to close the lane."*
3. Level falling. *"The sensor will turn it green on its own. I close the job when Team A reports clear."*
4. USB pull. *"I have lost the node. It goes grey, not green. I do not guess."*

6. Eric: the drain, the box, and the sensor

Sources: docs/Research/TabaPort_eric.docx (Drainage Sensor Placement for Urban Flood Monitoring). docs/Research/CalculationTabaPort.docx. docs/phase3/wiring.md. docs/phase3/bom.md.

6.1 What kind of drain, and where a sensor can sit

KCCA drains are main and second channels, community drains, road crossings and box culverts. Size at one site must come from drawings or a tape, not from "a standard Kampala drain." Eric note opens with the structures.



Figure E1. Box culvert (from Eric placement note). Sensor place: on the ceiling slab or headwall, **upstream** of the inlet splash, not in the silt on the floor.



Figure E2. Pipe culvert. Uganda guide in the note: at least **600 mm** under a road. Sensor sees water at an open chamber upstream, not inside the pipe.



Figure E3. Corrugated metal pipe. Same job as a concrete pipe. Mount and blind zone rules do not change.



Figure E4. Plastic pipe. Community drains often use these. Still measure width, floor and trash before you pick a mount.

Situation	Likely structure	Sensor sees water	Trash and silt
Roadside runoff	Open rectangle or trap channel	Stable section upstream	Litter at low points
Community drain	Small channel	Protected wall or overhead	Household waste, plants
Road crossing	**Box culvert** (Fig E1)	Upstream of the crossing, clear of splash	Debris at inlet, silt on floor
Smaller crossing	Pipe culvert (Figs E2 to E4)	Chamber upstream	Block at inlet
Main channel	Main or second channel	Stable section with a known height	Large debris, erosion

Uganda road drain guide in Eric note: pipe culverts under roads at least **600 mm**. Precast box often **600 mm by 450 mm** (sometimes 450 mm by 400 mm if the flow scours itself). Kampala tables go larger (pipes 750 to 1200 mm. Boxes 1 to 6 m span). **Survey the site.**

First build rule. An old, stable box culvert or open channel at a **known** flood spot. Not a section about to be rebuilt (Trevor then ranked the rebuilt Nakivubo at St Balikuddembe **low** for a first sensor).

6.2 Three ways to sense (Eric comparison drawing)

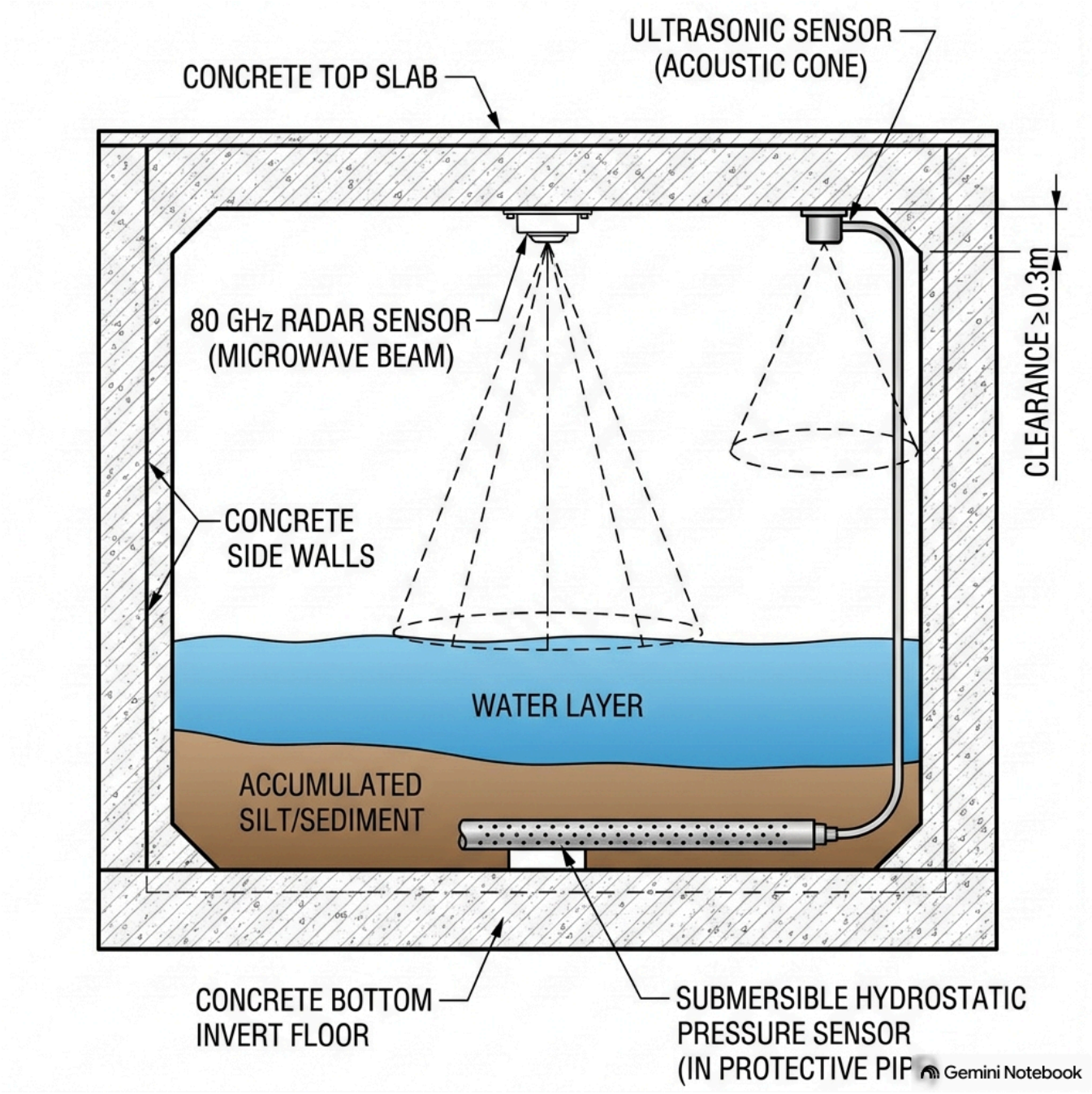


Figure E5. Eric comparison. Radar and ultrasonic look down at the water. Pressure sits in the silt. That is why the note rejects it for silt watch.

Way	Strength	Weakness in a Kampala drain	First build
80 GHz radar	Foam, steam, splash. About 2 mm. Tracks silt when dry by a shorter air gap	Cost. Shop wait	**Later** (Eric field pick)
Ultrasonic (JSN SR04T)	Cheap. No contact. Same time of flight idea	Air heat. Foam. 25 cm blind zone. Wet face	**What we bought and filmed**
Pressure	Direct depth from the bed	Buries in silt. Face clogs	Not for silt watch

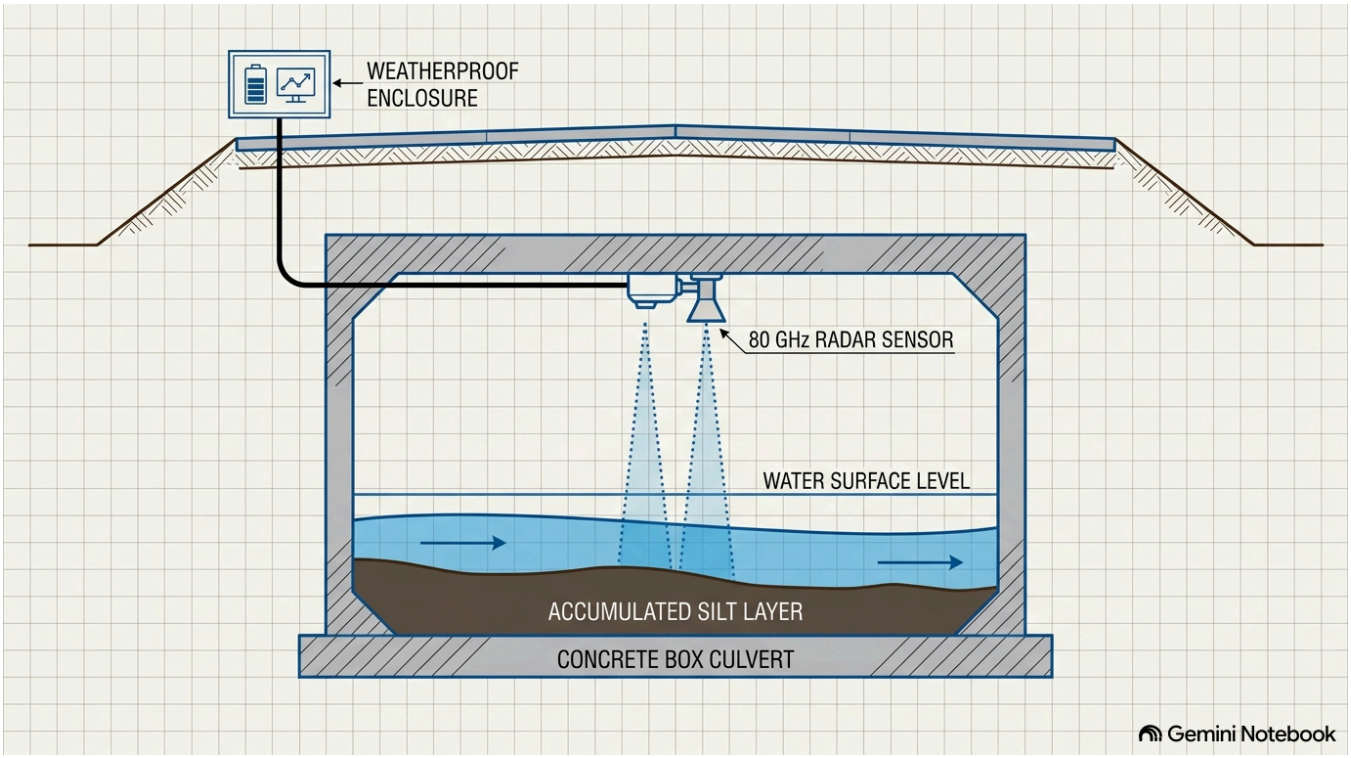


Figure E6. Radar only section from the placement note. This is the **field** pick, not what we filmed.

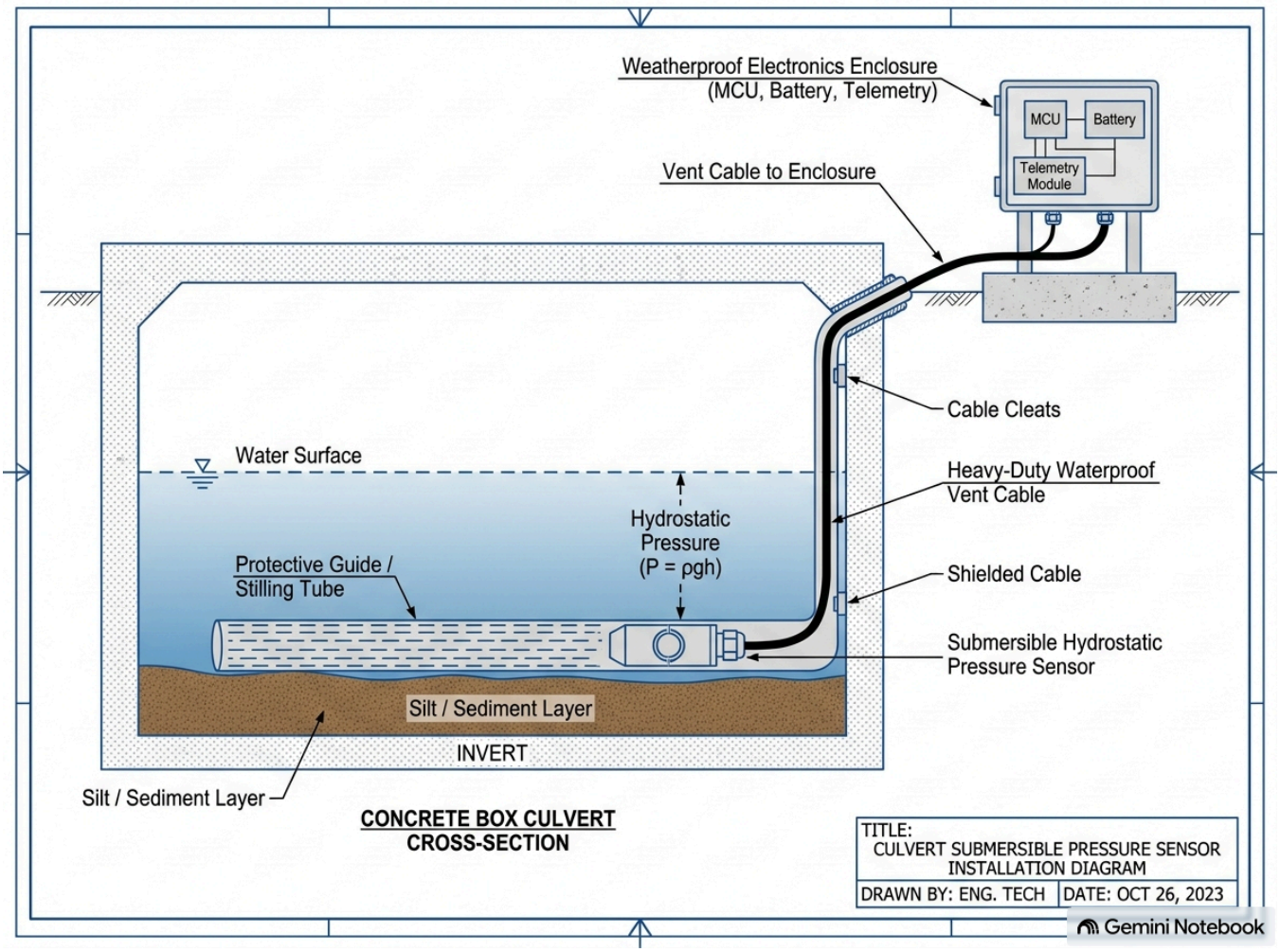


Figure E7. Pressure install. Eric: sand and mud bury the face. Drift and failure follow.

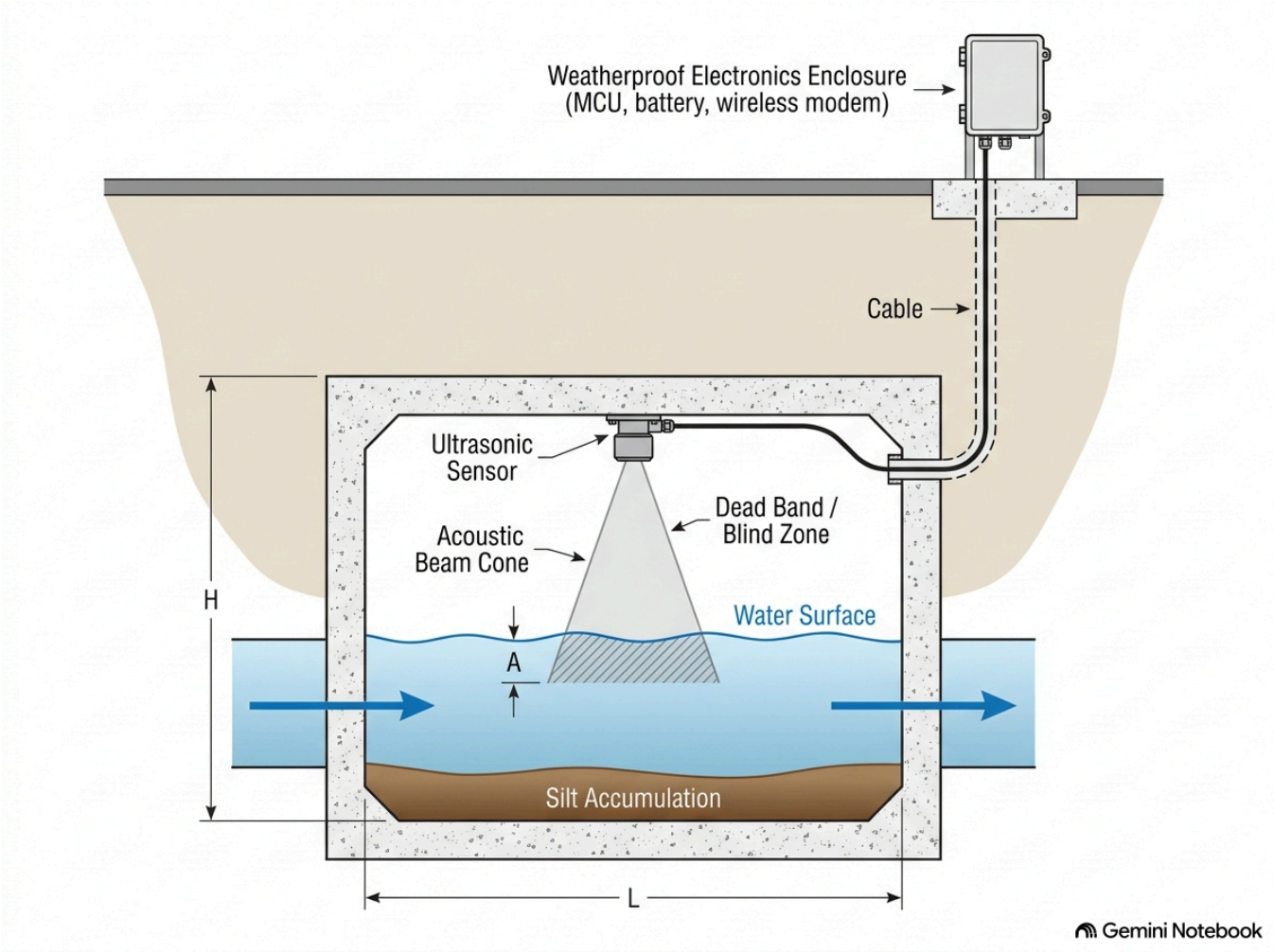


Figure E8. Ultrasonic (what the first build uses). Blind zone **A** must stay above max water. Height **H** and length **L** are taped on site.

Eric field pick is radar. The team still chose ultrasonic for Phase 3 because Neriko could sell a JSN SR04T the same morning (about UGX 25000) and the IEEE date would not wait on radar. If ultrasonic stays: stilling tube, 30 cm from a rough wall, 30 cm above max water, heat later.

Where the sensor must not go

- On the floor (silt). See Figure E7.
- In the inlet splash (branches, plastic).
- In the centre of the flow (desilt shovel).
- On a drain about to be dug up.
- In an obvious box that invites theft.

6.3 The maths from the calculation file

These drawings are from docs/Research/CalculationTabaPort.docx. They are the same formulae the hub and firmware use.

Distance with no contact (radar or ultrasonic):

d_measured = (v times t) / 2

Firmware (SRS): raw_cm = (echo microseconds times 0.0343) / 2. Speed of sound about 343 m/s.

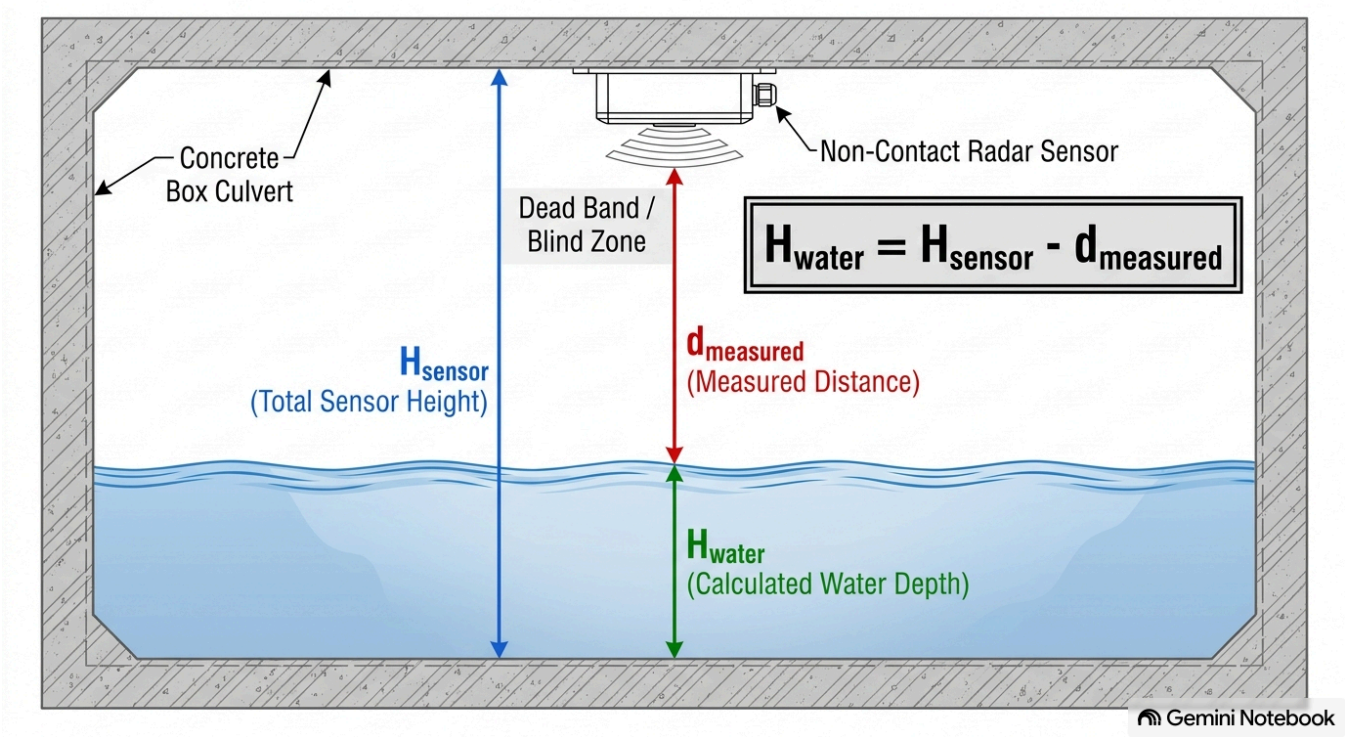


Figure E9. Calculation note. H_sensor is the dry tape from probe face to floor. d_measured is the live echo. The laptop and the node both work out H_water. The hub numbers are the source of truth.

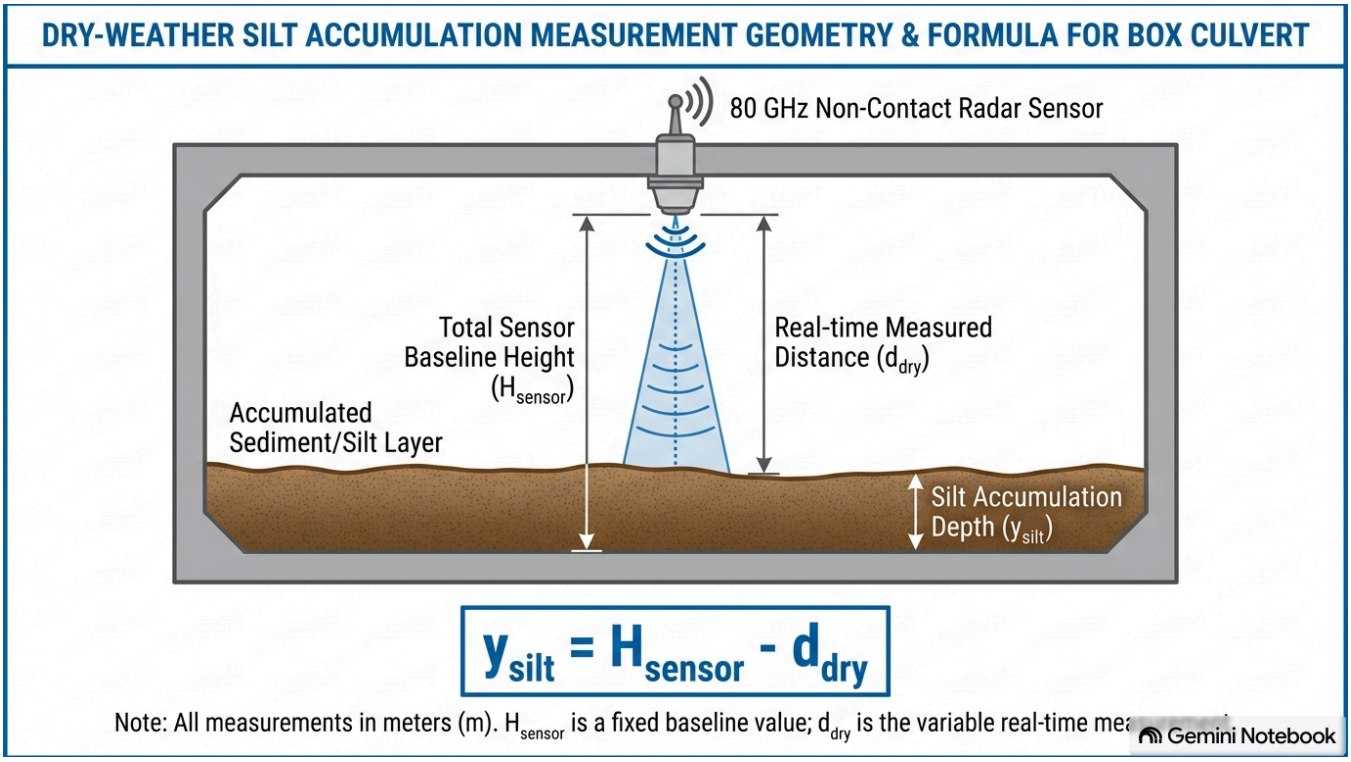


Figure E10. Same shape, dry channel. A lasting shorter air gap is silt on the bed, not a flood. Second build check. First build still needs a clean floor so H_sensor is true.

Blind zone: radar a few centimetres. Ultrasonic often **5 to 25 cm** (JSN SR04T about 25 cm). The probe must sit high enough that max flood never enters that zone.

Speed of sound shifts about **0.6 m/s per degree C** (about 1 cm error per 10 C at 50 cm). The field node design adds a heat sensor. The first USB node does not.

6.4 Two sensors for a block, and the field system Eric wants

Measure water where it has a clear meaning. Do **not** put the main sensor in the trash pile. Watch the **effect** of a block from upstream of the inlet.

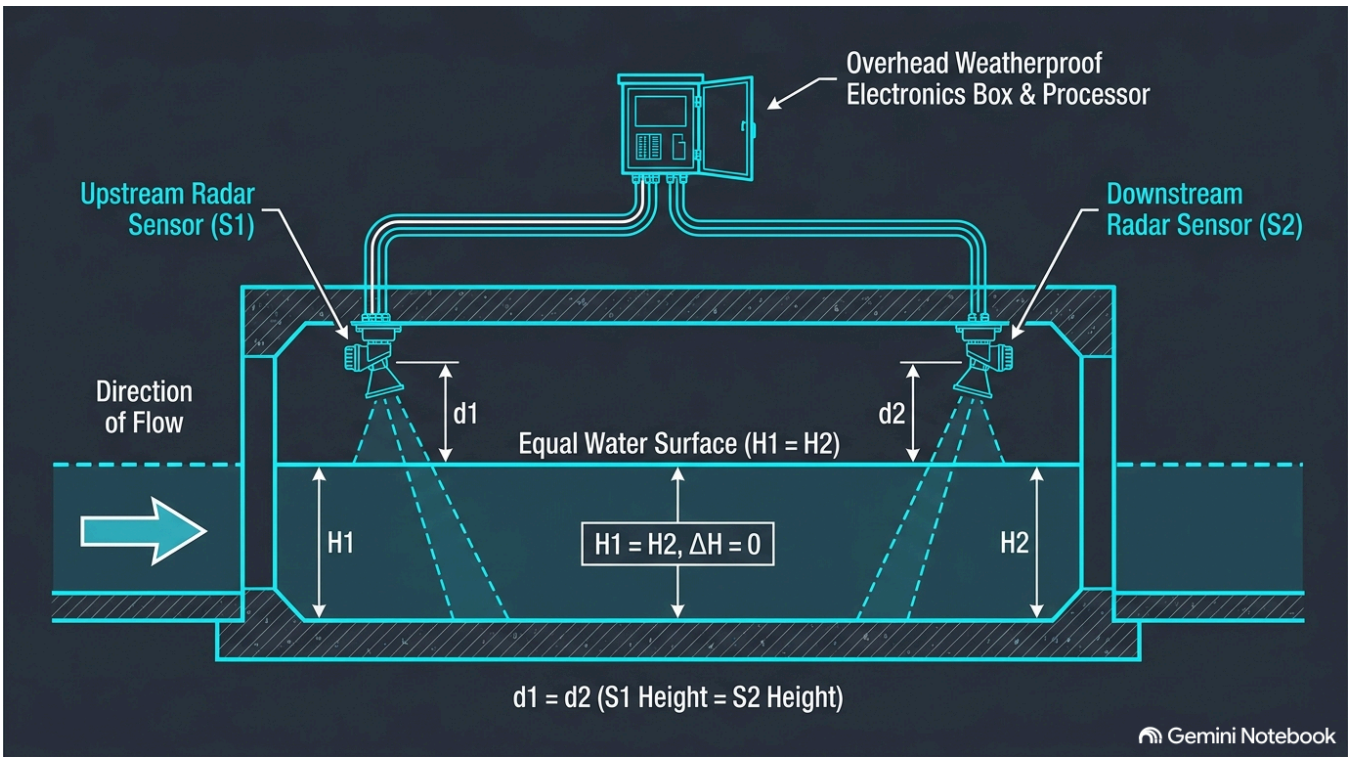


Figure E11. Block idea. Head gap (H upstream minus H downstream) rises when the crossing holds flow. Second build (two nodes). First build has one node.

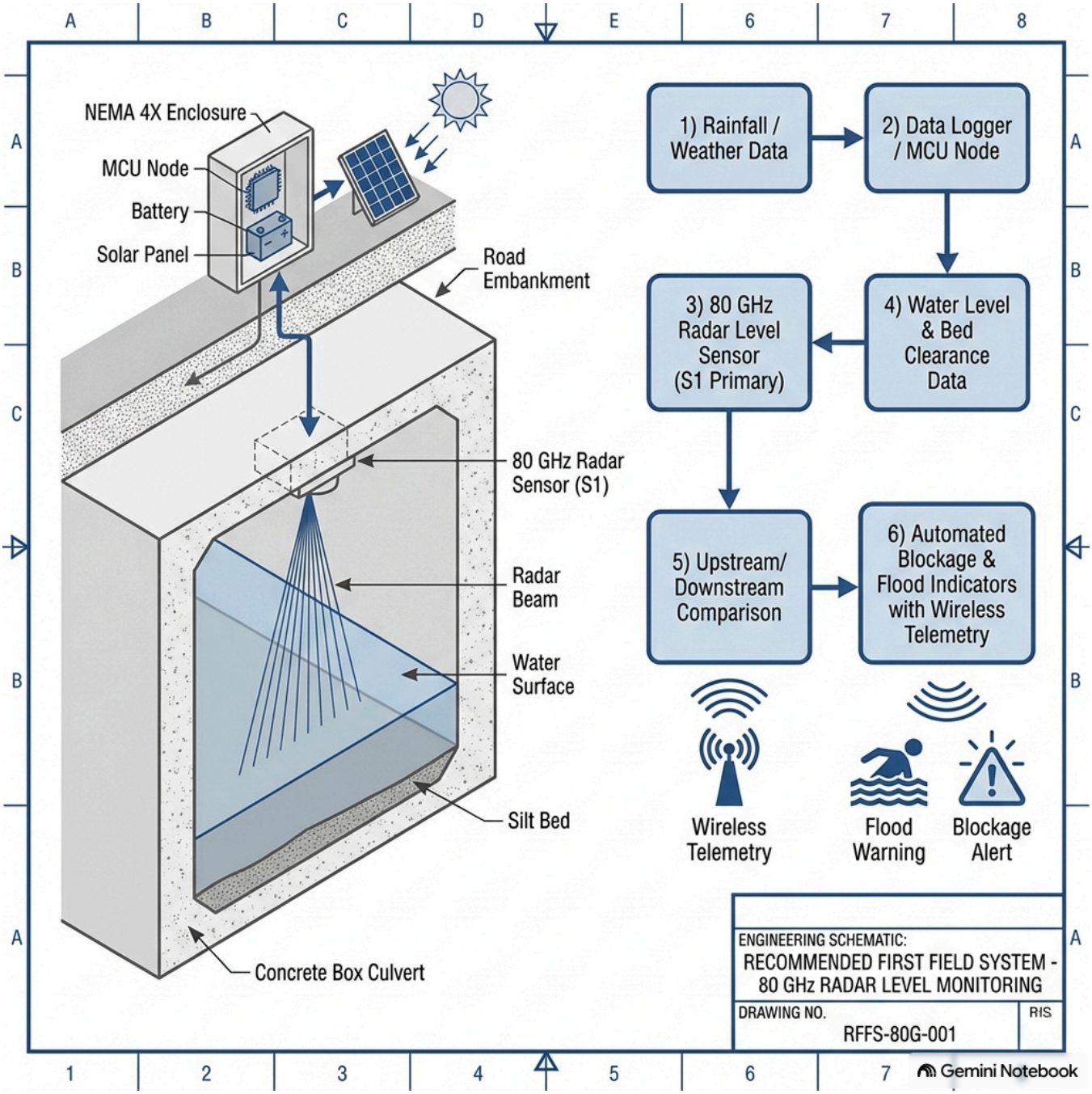


Figure E12. Field target after the IEEE video. Not what we filmed. First build is USB plus ultrasonic plus laptop.

6.5 How the sensor is fixed to the slab (calculation note)

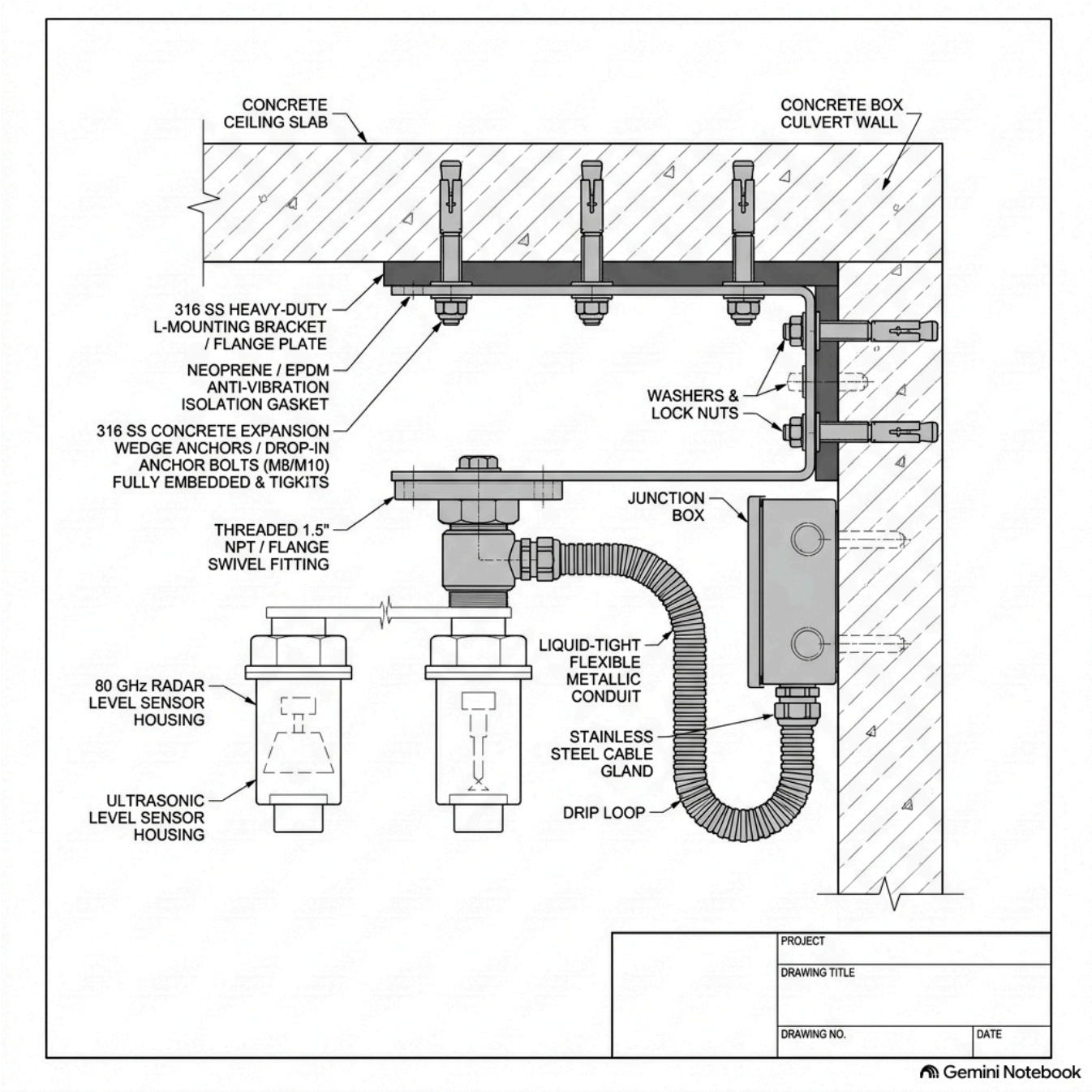


Figure E13. Calculation note. Suggested install. Probe straight at the water. At least 0.3 m from a rough wall. Downstream of the inlet splash. Blind zone clear of max flood. Cable in conduit with a drip loop **before** the box.



Figure E14. Wedge anchors from the calculation note (that file labels this "Figure 1: M8/M10 wedge expansion anchors"). First build used a plank and cable ties across a small campus drain. This drawing is the public culvert mount, after permission.

6.6 Threats and what we do

Threat	What happens	What we did or plan
Splash and waves	Unstable ultrasonic	Away from inlet. Plank mount. Stilling tube later
Trash	Spike or strike	Median of 5 plus 10 cm gate (Phase 2 filter, live)
Silt	Pressure sensor dies. Floor changes	Overhead only. Dry y_silt later
Sun and UV	Box dies	Lunch box for the video. Outdoor box later
Theft	Whole node gone	Bank box. Short film. No public install
Desilt	Crew hits the probe	Outside the shovel path. Talk to the crew
Building work	Shape changes	Eric and Trevor: skip rebuilt sections

KCCA already clears silt by hand and machine and works with communities. The box has to survive **maintenance**, not only rain.

6.7 First build wiring (what Eric and Shakiran built)

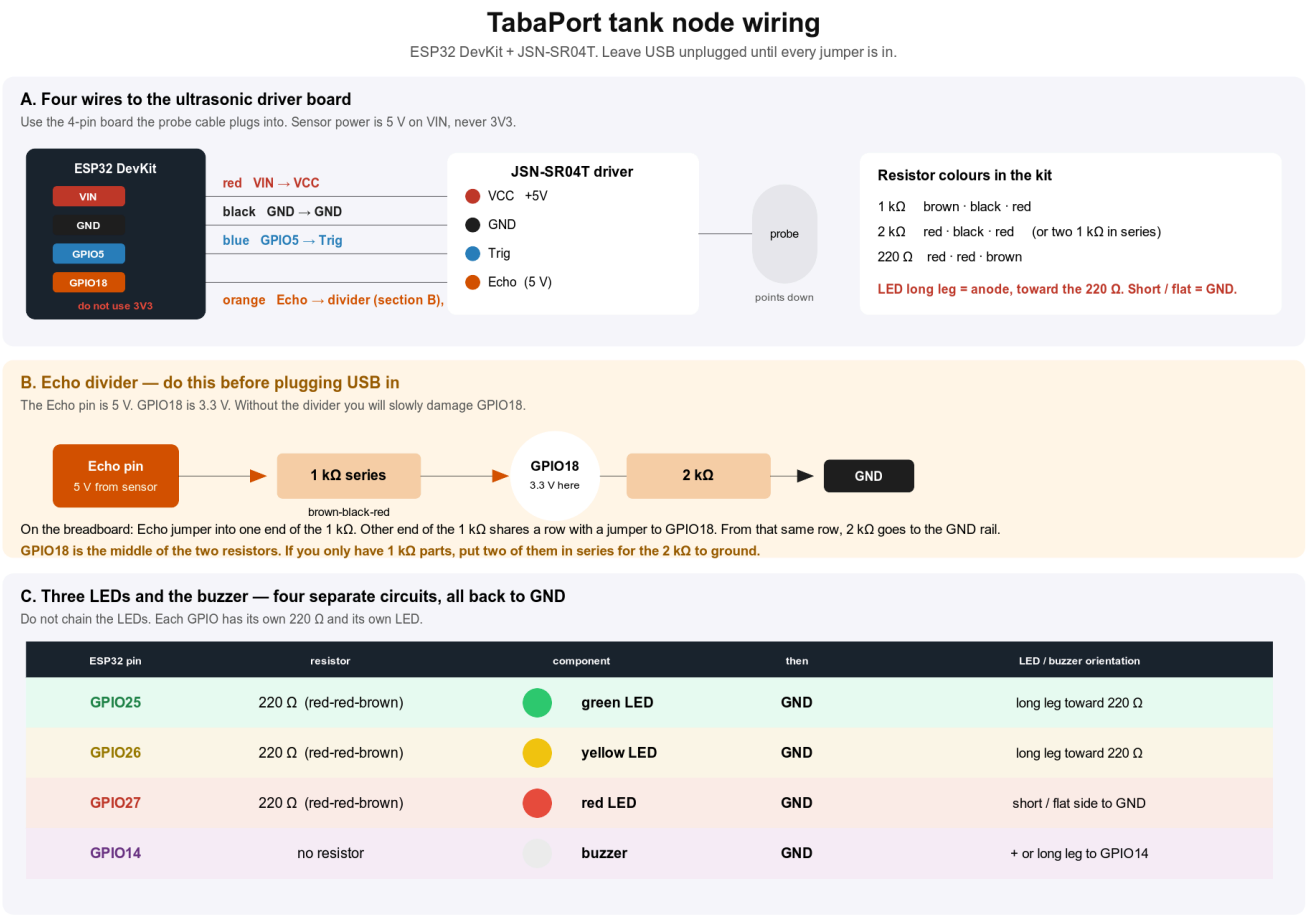


Figure E15. Printable wiring for the video node. VIN (5 V) to the JSN SR04T, **not** 3V3. Echo is 5 V. 1 k ohm and 2 k ohm divider into GPIO 18.

Function	ESP32 pin
Sensor power	VIN (5 V from USB)
Trig	GPIO 5. 20 us every 500 ms
Echo	GPIO 18 via divider
Green, yellow, red LED	GPIO 25, 26, 27 plus 220 ohm
Buzzer	GPIO 14. One beep on red

Mount rules (Eric, used on film day): face straight down (within 5 degrees). At least 30 cm above highest water. At least 30 cm from a rough wall. Plank weighted. Electronics in the lidded box. USB taped so feet do not pull it, unless we pull it **on purpose** for the unknown shot.

Parts (Neriko, Wandegeya, October 2026, plus or minus 20 percent): two ESP32 about UGX 120000. JSN SR04T about 25000. USB C **data** cable about 20000. Tape, box, plank about 33000. **About UGX 198000** with a spare board. **About 138000** with one. LEDs, buzzer, breadboard, resistors from the Arduino kit. Not buying: radar, LoRa, GSM, solar.

7. Mugisha Trevour Jeanclaude: where, and which map is true enough

Source: docs/Research/TabaPort_GIS_Component_Report_to_MIIC_Mugisha_Trevour.pdf (28 September 2026).
Code: TabaPort_GIS_Code/ (AHP, Earth Engine notes).

Maps answer one word for the whole product: **where.** A flood is a **stretch of road**, not a pin. The hub colours a segment_id, not one lat lon.

7.1 What the map work delivered

- A list of **17** known flood places across **three valley systems**.
- A flood score in Google Earth Engine on open data (AHP weights, consistency ratio **0.007**).
- A **short list of five** first sensor sites, led by Mulago roundabout and Kalerwe.
- A survey method that turns each echo into a water **height** on one datum (for a later trial).
- Every map file the laptop needs, made before the rain.

The centimetre shape is already drawn in Eric calculation figures (E9, E10, E13). A later trial still needs GNSS so those centimetres become heights. First build uses centimetres on one drain.

7.2 The 17 place list

Points are Wikipedia place (W) or author guess (E, about 500 m). **Guess until GNSS.** Basins wait until flow direction is confirmed.

ID	Place	Division	Basin (guess)	Lat, Lon	Why it floods	Trial
TP 01	Clock Tower junction (Entebbe Rd / Queensway)	Central	Nakivubo	0.3095, 32.5785	Valley floor below Nakasero. Roads meet on one low crossing	High
TP 02	Bwaise	Kawempe	Lubigi via Nsooba	0.3500, 32.5611	Old wetland about 1180 m. Dense homes on the drain	High
TP 03	Kalerwe market (Bombo / Gayaza)	Kawempe	Lubigi via Nsooba	0.3525, 32.5720	Market on the Nsooba line. Heavy waste into drains	High
TP 04	Mulago roundabout	Kawempe	Lubigi via Nsooba	0.3395, 32.5745	Dip at the foot of Mulago Hill beside the national hospital	High
TP 05	Kubiri roundabout (Bombo Rd)	Kawempe	Lubigi via Nsooba	0.3460, 32.5690	Low junction Wandegeya to Bwaise	Medium
TP 06	Northern Bypass at Lubigi wetland	Kawempe / Rubaga	Lubigi	0.3470, 32.5500	Highway on the wetland floor. Not high ground **here**	High
TP 07	Nakivubo at St Balikuddembe	Central	Nakivubo	0.3110, 32.5775	Crowded CBD channel. Rebuilt to 12 m by March 2026	**Low** (rebuilt)
TP 08	Lugogo Bypass	Nakawa / Central	Nakivubo (Lugogo)	0.3285, 32.5990	Low road along the Lugogo feeder	Medium
TP 09	Mukwano Rd near Kibuli	Central / Makindye	Nakivubo	0.3070, 32.5930	Industry valley floor	Medium

ID	Place	Division	Basin (guess)	Lat, Lon	Why it floods	Trial
TP 10	Nateete (Old Masaka Rd)	Rubaga	Nalukolongo	0.2983, 32.5350	Valley road about 1190 m on Nalukolongo head waters	Medium
TP 11	Nalukolongo channel	Rubaga	Nalukolongo	0.2960, 32.5490	Channel reported too small (Parliament July 2026)	High
TP 12	Ndeeba	Rubaga	Nalukolongo	0.2940, 32.5610	Low shops, Masaka Rd	Medium
TP 13	Kinawataka	Nakawa	Kinawataka	0.3350, 32.6330	Swamp floor, main channel	Medium
TP 14	Kyambogo	Nakawa	Kinawataka	0.3480, 32.6290	Low ground below Kyambogo hill	Medium
TP 15	Ntinda (factory close)	Nakawa	to confirm	0.3550, 32.6144	Valley between Ntinda and Naguru	Medium
TP 16	Kansanga / Ggaba	Makindye	Kansanga	0.2872, 32.6078	Kansanga swamp line	Medium
TP 17	Port Bell	Nakawa	Lake shore	0.2890, 32.6530	About 1135 m. Lake back water	Low

Three valley systems

1. **North (Lubigi / Nsooba):** Bwaise, Kalerwe, Kubiri, Mulago, Northern Bypass.
2. **Centre (Nakivubo):** Clock Tower, Nakivubo, Mukwano, Lugogo.
3. **South west (Nalukolongo):** Nateete, Nalukolongo, Ndeeba.

A natural sensor chain. Mulago to Kubiri to Kalerwe to Bwaise look like one line. If flow direction confirms it, an upstream node gives minutes of warning down stream. From "where is water now" to "where will water be next."

First sensor short list (site check and KCCA permission still required):

1. TP 04 Mulago roundabout (hospital access, top of the north chain)
2. TP 03 Kalerwe (named in every team file)
3. TP 01 Clock Tower (headline flood spot. Check current KCCA works)
4. TP 11 Nalukolongo (channel too small, Parliament July 2026)
5. TP 02 Bwaise (down stream end of the chain)

Place pick follows Eric placement note (Figs E1, E5, E13): old stable structure, not a rebuild, not the trash zone.

7.3 Flood score (Google Earth Engine)

The model scores every **30 m** cell of Kampala from 0 (dry ridge) to 1 (valley floor). Open Earth Engine data only.

Factor	Data	Direction	Weight
Height above nearest drain (HAND)	MERIT Hydro v1.0.1	Lower = more at risk	**0.316**
Low point (local hollow)	Copernicus DEM GLO 30	Deeper hollow = more at risk	0.197
Wetness index	MERIT Hydro	Wetter = more at risk	0.184

Factor	Data	Direction	Weight
Distance to drain	MERIT Hydro	Nearer = more at risk	0.107
Hard ground	Dynamic World V1	More sealed = more at risk	0.107
Slope	Copernicus DEM GLO 30	Flatter = more at risk	0.057
Rain extremes	CHIRPS Daily, 1995 to 2025	Wetter = more at risk	0.032
Consistency ratio	Saaty method	Limit 0.10	**0.007**

HAND and hollows carry the weight because, across 15 km, **rain differs little** while height above the drain differs a lot. Coded in TabaPort_GIS_Code/ahp.py (static score plus the three IEEE rank weights: rise rate, key buildings, people).

Honest limit on satellites. Sentinel 1 before and after March 2025, October 2025 and July 2026 can map **standing** water. Radar repeats every 6 to 12 days and fails between tall buildings, so it **misses Kampala flash floods that drain in hours**. That limit **is** the case for TabaPort: space cannot see the event. A probe in the culvert can.

Test written in the GIS report: at least **12 of 17** list points in the top two classes.

7.4 Map files with no internet (what Sharif hub loads)

All map work happens **before** the rain. The laptop needs no GIS server, no cloud tiles, no API key. Sharif stack research picked Leaflet, local tiles and GeoJSON for the same reason.

File	Contents
road_segments.geojson	Roads cut into segments with IDs and flood fields
blackspots.geojson	TP 01 to TP 17
Local tiles	Context under the segments
Gazetteer	Place names and aliases to a segment plus a 100 m circle

Human reports have no GPS. The gazetteer snaps *"Kalerwe mkt flooded"* to TP 03 / the Kalerwe segment.

Phase 2 named the Northern Bypass as high ground. Trevor list marks the **Lubigi stretch of that Bypass (TP 06) as a flood spot**. So "take the Bypass" is not a blanket rule. High ground is measured per segment, not assumed from the road name.

7.5 Map limits we state

- A 30 m height model cannot see a kerb or a one metre dip. The score points the survey. It is not a water model.
- 13 of 17 points are guesses until GNSS.
- OSM is thin in Bwaise and Kalerwe (YouthMappers map days later).
- No field install without written KCCA permission.

8. Sharif: stack, data, job flow, and the SRS

Sources: docs/Research/Sharif_Research.docx (Stack and Data, April 2026). docs/SRS.md and docs/TabaPort_SRS.docx.

Question the software research answered: *what can we honestly code by the deadline, and how does it stay offline when the phone network fails?*

8.1 Map options with no internet

Option	Strength	Why it waited or won
Static PNG only	Zero code	No pan, zoom, or click
SpatiaLite	Real space SQL	Extra install. Too much for 69 points. Phase 2 named it. We did the map work **before** the rain instead
MapLibre GL	Modern vectors	Needs a tile pipe we did not have
PWA	Right long term form	Extra work for a laptop desk
Leaflet, local tiles, GeoJSON	Small core. Segments you can edit by hand. leaflet.offline	**Won**

The laptop runs Flask. Leaflet, tiles and GeoJSON live on disk. Dispatcher opens http://localhost:5000. No CDN. No key.

8.2 Data model (four tables, built as specified)

Everything in one SQLite file, WAL mode. Nothing personal goes to a cloud.

node_reading: reading_id, node_id, time, water_level_cm, rise rate, filtered_flag, raw ultrasonic.

road_segment_state: segment_id, optional node_id, state (green, yellow, red, unknown), last_updated, source (sensor, human_report, dispatcher).

job: job_id, segment_id, optional alert_id, assigned_crew, status (open, acked, en_route, on_site, closed, false_alarm), times, false_alarm_reason.

human_report: type (blockage, flooding, road_impassable, clear), note, **reporter_hash only**.

Privacy: salted SHA 256 of the phone number at the gateway. Uganda Data Protection and Privacy Act treats phone numbers as personal data. UCC SMS guide flags privacy. Raw numbers are not in SQLite or CSV.

8.3 Node to hub

Link	Range and rate	Verdict
USB serial and PySerial	About 5 m, 115200, one cable	**First build**
WiFi / MQTT	Wireless, broker, reconnect	Second build (field node sim uses MQTT as a stand in)
LoRa	Private, needs a gateway	Research only

Link	Range and rate	Verdict
GSM SMS	Works where the cell works. Costs. Jams in storms	Human reports later, not the node

JSON line, new line at the end, json.loads on the laptop:

```
{"node_id":"drain_01","raw_cm":41.2,"level_cm":8.8,"filtered":false,"state":"green","ok":true}
```

Hub is source of truth. CFG back to the node:

```
CFG h_sensor=60.0 yellow=30.0 red=48.0
```

8.4 After the alert (software)

A **job** is the work order. An **alert** is the signal. One alert makes one job. A human report can open a job with no sensor.

Step	What happens	Job status
1	Node at Clock Tower reads 25 cm, rising 6 cm per min	
2	Hub creates alert and job	open
3	Dispatcher assigns Team A	open (owned)
4	Team A on radio. Dispatcher Ack	acked
5	Crew arrives, clears inlet	en route, then on site
6	Sensor reads 8 cm. Close	closed
7	Segment green **because the sensor said so**	

If step 5 is only a log: False Alarm, reason logged, filter tuned later. With no clear false alarm path, dispatchers stop trusting the board.

Job list: unacked first. One click Assign, Ack, Close, False Alarm. No confirm popups.

8.5 Uganda SMS and USSD, and why cell broadcast is not ours

Option	Cost (Sharif 2026 research)	Notes
Shared shortcode plus keyword	About UGX 200000 setup	Hours after KYC, not months
Reverse billing in	About UGX 65 per message	Rider pays nothing
Bulk SMS out	About UGX 30 to 55 per message	Volume and sender ID
Dedicated shortcode	UCC USD 295 plus USD 2360 a year plus telco setup	About 2 months. Not first build

Cell broadcast is out of scope for three reasons Sharif wrote down:

1. UCC treats emergency area alerts as a **government** function, not a student board service.
2. There is no Cell Broadcast Centre we can buy.

3. Cell broadcast is one way. We need a rider to **send** a report.

Phase 2 "zero cost SMS to every phone in 2 km" stays in the idea as a **UCC, MTN, Airtel partnership**. The video does not pretend to do it.

8.6 SRS lock (October 2026)

Sharif wrote the SRS. It is the build contract: JSON and CFG, REST, tables, tests AC 01 to AC 12.

In scope: one USB node, laptop hub, Leaflet with no internet, SQLite, Flask, four states, job life, typed reports, USB pull unknown demo.

Out of scope: LoRa and GSM, Pi hub, cloud tiles or DB, cell broadcast, KCCA system join, auto USSD, solar.

One reading (SRS 3.2): TRIG, then echo, then firmware median, then level_cm, then JSON, then PySerial, then hub median, then hub calibration, then class, then SQLite, then /api/state every 2 s, then Leaflet and jobs table.

9. What we built: the first prototype (with the screens)

Folder: prototype/app (hub) and prototype/sysmatic/tank_node (firmware).

Firmware (Shakiran): ESP IDF, 2 Hz, median of 5, 10 cm spike gate, JSON, LEDs, beep on red. The laptop **works out depth again** from the raw echo so a wrong setting on the node cannot lie to the map.

9.1 Live node

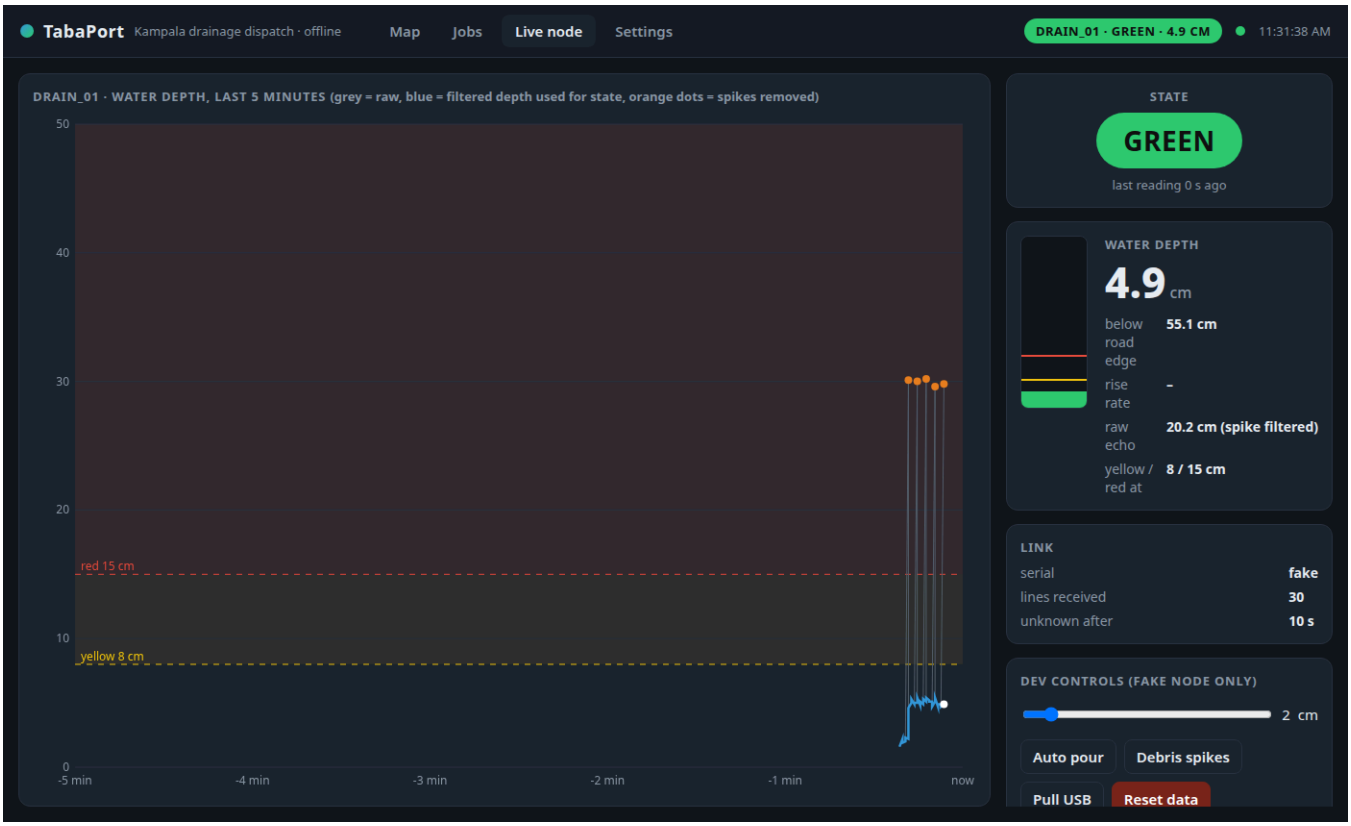


Figure 12. Live node. Grey is raw. Blue is filtered depth used for state. Orange dots are spikes removed (the Phase 2 plastic bottle filter, on screen). Yellow and red lines are Joel centimetres. Fake serial slider in this shot is for rehearsal. The IEEE tape uses the real USB node.

9.2 Map

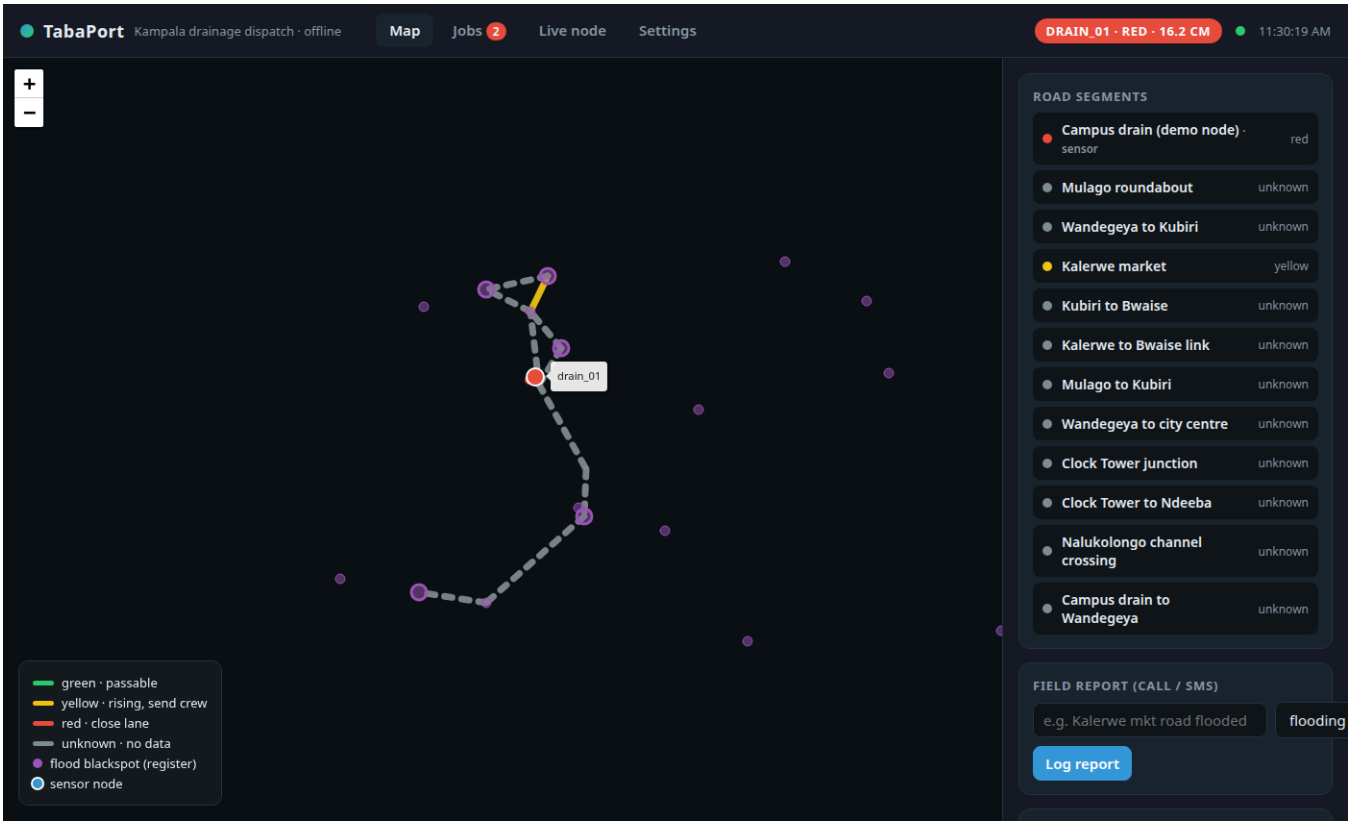


Figure 13. Map. Campus drain (drain_01) red. Purple dots are Trevor list (Clock Tower, Kalerwe, Bwaise, Mulago, Nalukolongo). Grey is unknown (no node). Field report box: a typed "Kalerwe mkt flooded" snaps through the gazetteer. Legend: green passable, yellow send crew, red close lane, grey no data.

9.3 Jobs

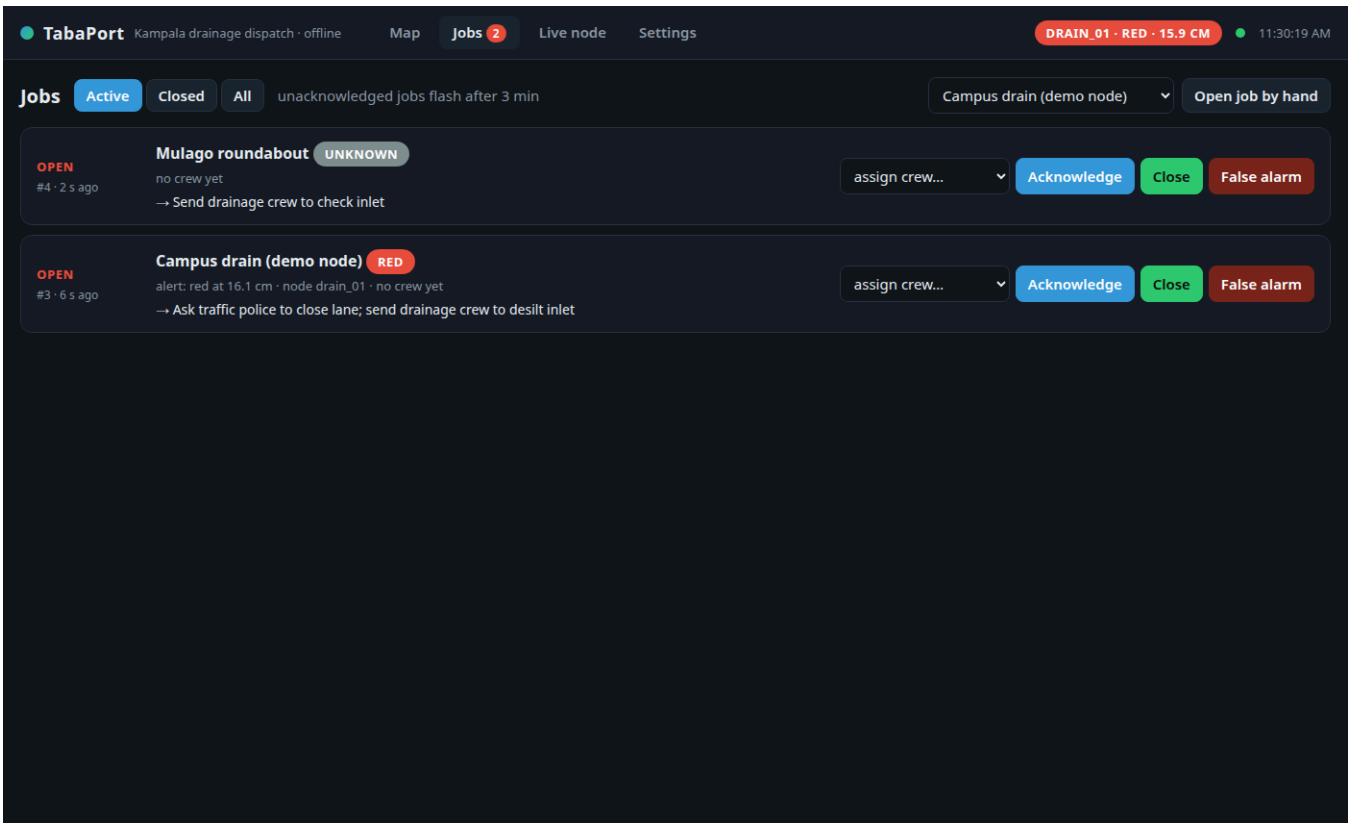


Figure 14. Jobs. Open job 3: campus drain **red** at 16.1 cm. *"Ask traffic police to close lane. Send drainage crew to clear inlet."* Open job 4: Mulago roundabout **unknown**. *"Send drainage crew to check inlet."* Unacked rows flash after 3 minutes. Buttons: assign, acknowledge, close, false alarm.

That is Joel language on Sharif table on Trevor place names, driven by Eric and Shakiran node.

10. Shakiran: firmware, join, IEEE, phone

After Phase 2 the lead job was not a new slogan. It was: keep the team on **one** first build slice, write firmware a real drain can drive, and put a person on the product.

What we will not show in the IEEE video: Wokwi, slides, a fake graph sold as a drain, cell broadcast, a city scale dashboard.

Phone app (extra). Flutter client, same REST API (prototype/mobile/). A physical phone starts at the public hub URL. The Linux desk stays on 127.0.0.1:5000. The Phase 3 tape is the **laptop**.

Field node (sim only). prototype/sysmatic/field_node. Heat sensor, battery, tilt, clock, LCD, microSD queue, MQTT stand in. Questions for a WSN teacher (power, LoRa versus GSM, box, lightning) are in docs/phase3/field-node-brief.md. **Not Phase 3 proof.**

11. The Phase 3 video

Rule: a real person, a working product, 2 to 5 minutes. No sim. No stock. No AI video.

Role on the day	Person
Drain, probe, crew	Eric
Laptop, dispatcher	Shakiran
Camera 1 and site	Trevor
Water, timing, second camera	Joel
Hub standby	Sharif

Shot list (aim about 3 min 30): wide drain, plank, box. Live node green, WiFi off. Pour to yellow, assign, ack. Pour to red, beep, map red. Crew clears inlet, sensor turns green, close. USB pull, grey unknown, plug back. Optional gazetteer and route that skips red. Team end card with the **five names**.

12. What we are not claiming

- That TabaPort stops flooding.
- That ultrasonic in a dirty **public** culvert is already good enough for live KCCA dispatch. That needs a 30 day trial beside patrol reports.
- Cell broadcast to every phone in 2 km.
- A public drain install. KCCA asset. Written permission first.
- That the Phase 2 US$55 solar radio node is what we filmed. We filmed a USB node.
- That we have already sat with a KCCA drainage engineer. We have not.
- That the 30 m score surface is a water model.

13. Next

1. Keep the first build as the SRS reference.
2. Ask Dr Mary Nsabagwa, Makerere CoCIS (embedded and WSN, WIMEA ICT weather stations) to mark the **field** node. Read Nsabagwa et al. 2019 and 2018 condition watch first. Brief: docs/phase3/field-node-brief.md. Her four tests: reliability, accuracy, power, send. **Cost after those.**
3. Then a 20 minute KCCA Engineering **talk** (not a money ask): how reports reach crews today. Which 5 of 69 first. Whether a gentle demo on their land is ok.
4. One real flood spot, 30 days, side by side with patrols.

14. Files to send with the video

- This report as PDF
- SRS: docs/TabaPort_SRS.docx or print of docs/SRS.md
- Phase 2 PDF: docs/phase2/QM_Idea_5267_08_23_2026.pdf
- Research originals
- docs/Research/Sharif_Research.docx
- docs/Research/TabaPort_eric.docx
- docs/Research/CalculationTabaPort.docx
- docs/Research/TabaPort_GIS_Component_Report_to_MIIC_Mugisha_Trevour.pdf
- docs/phase3/thresholds-and-words.md
- docs/phase3/wiring.md and docs/phase3/bom.md
- IEEE upload confirm screen shot
- Team photo or end card with the five names

15. Combined references

Team files

1. TabaPort team (entrant: Shakiran Nannyombi). Phase 2 idea. Tabaport: On Site Edge Intelligence and Dispatch System for Urban Flash Floods. IEEE Response Quest 5267, 24 May 2026. docs/phase2/QM_Idea_5267_08_23_2026.pdf
2. Sharif. Software Requirements Specification, TabaPort first build. October 2026. docs/SRS.md
3. Sharif. TabaPort Research Document: Stack and Data. April 2026. docs/Research/Sharif_Research.docx
4. Eric. Drainage Sensor Placement for Urban Flood Monitoring. docs/Research/TabaPort_eric.docx
5. Eric. Time of flight, H_water, y_silt, blind zone and mount. docs/Research/CalculationTabaPort.docx
6. Mugisha Trevour Jeanclaude. TabaPort: GIS and Geomatics Component. Report to MIIC TES, 28 September 2026. docs/Research/TabaPort_GIS_Component_Report_to_MIIC_Mugisha_Trevour.pdf
7. Joel. Thresholds, job wording and the dispatcher lines. docs/phase3/thresholds-and-words.md
8. TabaPort team. MIIC TES concept note. docs/miic-tes-concept-note.md
9. TabaPort team. Research week brief. docs/phase3/team-research-week.md
10. TabaPort team. Wiring, BOM, shot list. docs/phase3/wiring.md, bom.md, rehearsal-and-shotlist.md

News and agencies (posts we used)

11. KCCA. Drain and flood ready update (waste and silt). https://www.kcca.go.ug/news/1154/
12. UBC. KCCA steps up drainage works to tackle flooding in Kampala. 12 August 2026. https://ubc.go.ug/2026/08/12/kcca-steps-up-drainage-works-to-tackle-flooding-in-kampala/

13. Daily Monitor. Parliament gives KCCA 7 days to explain Kampala flooding. https://www.monitor.co.ug/uganda/news/national/parliament-gives-kcca-7-days-to-explain-kampala-flooding-5543126
14. NTV. KCCA urges land donation to solve Kampala flooding. https://ntv.co.ug/news/kcca-urges-land-donation-to-solve-kampala-flooding
15. OpenStreetMap Uganda. https://www.openstreetmap.org
16. Uganda National Meteorological Authority (climate context, Phase 2 form).
17. KCCA Strategic Plan 2025/26 to 2029/30. Drain and flood spot programmes (Eric).
18. Ministry of Works and Transport. Road Design Manual Volume II: Drainage Design Manual. 2010 (culvert minima cited by Eric. Confirm the current official PDF before you quote to KCCA).

Technical papers

19. Nsabagwa, M., Byamukama, M., Kondela, E., and Otim, J. S. (2019). Towards a robust and affordable Automatic Weather Station. Development Engineering, 4, 100040. https://doi.org/10.1016/j.deveng.2018.100040
20. Nsabagwa, M., Sansa Otim, J., Nyongarwizi Akol, R., et al. (2018). Condition Monitoring for Wireless Sensor Network Based Automatic Weather Stations. EAI Endorsed Transactions on Internet of Things, 4(14), e4. https://publications.eai.eu/index.php/IoT/article/view/661
21. Toa, M., and Whitehead, A. (2021). Ultrasonic sensing basics (SLAA907D). Texas Instruments. (Eric)
22. Omega Engineering. LVCN210, LVCN318 and LVCN414 ultrasonic user guide, M4462. (Eric)
23. Hunan MAC Sensor. How to use, install and maintain ultrasonic level sensor. 2025. https://www.macsensor.com/info/how-to-use-install-and-maintain-ultrasonic-le-103166307.html
24. Icon Process Controls. Guide to radar liquid level sensors. 2025. https://iconprocon.com/blog_post/a-complete-guide-to-radar-liquid-level-sensors-precision-measurement-for-complex-applications/
25. JSN SR04T data sheet (blind zone, beam, cable. Use the sheet that came with the part).
26. Uganda Data Protection and Privacy Act. Phone numbers as personal data.
27. IEEE Response Quest Challenge rules. Real person, working product, 2 to 5 minutes.
28. Saaty AHP and Earth Engine inputs: MERIT Hydro, Copernicus DEM GLO 30, Dynamic World, CHIRPS (Trevor).

Sign off

We confirm this report describes research and a prototype the team actually made after the Phase 2 idea. KCCA and UNDP numbers are labelled as theirs. Yellow and red centimetres are **our** rule from site shape, not a published national flood table.

Name	Role	Date
Shakiran Nannyombi	Lead and firmware	7 October 2026
Joel	Bands and dispatch words	7 October 2026
Eric	Drain and hardware	7 October 2026
Mugisha Trevour Jeanclaude	Maps	7 October 2026
Sharif	Software and SRS	7 October 2026